The Missing Twins:
Transmission Channels Between Currency and Debt Crises

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Abstract

In this paper, I determine that currency crises help predict debt crises, and expand on existing literature investigating the role of foreign currency-denominated debt in currency and debt crises to propose some transmission channels that could help explain this association. Using a novel methodology combining a regression-based event study with a linear probability model, I find that foreign exchange reserve declines and tax revenue declines both (i) have a high likelihood of occurring following a currency crisis, and (ii) are associated with an increase in the likelihood of a debt crisis. Combining the coefficients from the models, I estimate that the fall in reserves associated with a currency crisis increases the likelihood of a debt crisis by up to 1.6%, while the decline in tax revenue associated with a currency crisis increases the likelihood of a debt crisis by up to 2.9%. These increases are non-trivial compared to the unconditional probability of a debt crisis, which is 17% in my sample. The empirical results thereby provide support for theories emphasizing the role of foreign exchange reserves and tax revenue in the transmission from a currency crisis to a debt crisis.

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1 Introduction

In this paper, I show that currency crises help predict debt crises, and that such joint crises are associated with larger GDP declines and higher inflation than currency-only or debt-only crises. While existing literature provides theories of how the two crises may be associated, I take a step further in using a novel methodology to empirically test possible transmission channels. I find that foreign exchange reserve declines and tax revenue declines are significant in the interaction between currency and debt crises. These findings provide empirical support for theories of how foreign currency reserves and tax revenue affect transmission from a currency crisis to a debt crisis through their connection with foreign currency-denominated debt.

Although many papers have been devoted to studying associations between different types of financial crises, the link between currency and debt crises has not received as much attention. However, many countries experiencing currency crises have also concurrently experienced debt crises. Recent examples of such crises include Mexico in 1994, Russia in 1998, and Argentina in 2001.

I thus begin my empirical work by examining whether currency and debt crises can help predict each other using a vector autoregression model (VAR), following the methodology of Reinhart and Rogoff (2010a). The model provides evidence that currency crises help predict debt crises, while debt crises help predict currency crises over a shorter time frame. I then establish that joint currency and debt crises are associated with larger declines in growth and higher inflation than either crisis individually via an OLS framework. The connection between joint currency and debt crises and more severe macroeconomic outcomes parallels similar analyses done for other twin crises (Kaminsky and Reinhart 1999 ; Reinhart and Rogoff 2010a), and indicates the importance of understanding how currency and debt crises are connected.

Having established that currency crises help predict debt crises, I describe and empirically test transmission channels that could help explain their connection. While the theoretical nature of transmission channels has been documented in existing literature, no paper has empirically tested these propositions. Therefore I begin by using existing theory to present some potential transmission channels between the two crises, and then empirically determine whether these channels are supported by the data.

I accomplish the empirical step in two parts. First, I utilize a regression-based event study to observe the evolution of foreign exchange reserves, tax revenue, government expenditure, and the current account balance during a currency crisis. I select these variables because I expect them to have an impact on the sustainability of the debt level, specifically through their connection with foreign currency-denominated debt. The study provides evidence that a currency crisis is associated with a decline in foreign exchange reserves and a decline in tax revenue, which is consistent with existing literature (Kaminsky, Lizondo and Reinhart 1998 ; Kaminsky and Reinhart 1999).

Second, I adapt the methodology of Bordo, Meissner, and Stuckler (2009) by constructing a linear probability model to test how foreign exchange reserves, tax revenue, government expenditure, and the current account balance are associated with the likelihood of a debt crisis. While this component of my analysis builds on previous work, the way in which I combine it with the regression-based event study is an original contribution. I find evidence that a decline in foreign exchange reserves and a decline in tax revenue are in turn associated with a higher likelihood of a debt crisis.

By cross-multiplying the coefficients from the two parts of the study, I am able to assess the significance of each channel in the transmission between currency and

debt crises. Although these estimates cannot isolate the exact effect of each channel on the transmission in the direction from currency crises to debt crises due to endogeneity issues, they can nevertheless provide evidence in support of the transmission channels described. In this way, I find evidence that foreign exchange reserves and tax revenue are significant in the transmission between currency and debt crises. I find no supporting evidence for transmission channels via government expenditure or via the current account balance.

The rest of the paper proceeds as follows. Section 2 reviews background literature on connections between currency and debt crises. Section 3 explains the transmission channels which I test. Section 4 summarizes the data sources and presents some summary statistics. Section 5 explains the empirical methods and specifications. Section 6 presents and discusses the empirical results. Section 7 concludes and provides avenues for future work.

2 Related Literature

Existing literature identifies several different types of financial crises, most commonly differentiated as currency, banking, debt, and inflation crises (Reinhart and Rogoff 2009). Empirically, it is found that different crises frequently occur concurrently or sequentially, and that such twin crises tend to be more severe than individual crises (Kaminsky and Reinhart 1999 ; Reinhart and Rogoff 2009).

The link between currency and debt crises has not been as widely analyzed as other twin crises in literature, despite the fact that they frequently occur together. Reinhart (2002) finds that 84% of debt crises in her sample were associated with currency crises, and 58% of currency crises were associated with debt crises. Similarly, Herz and Tong (2004) find that 32% of debt crises in their sample were linked to currency crises, and 20% of currency crises were linked to debt crises.[1] These statistics reflect the fact that debt crises are less common than currency crises, but also suggest that the two crises could be associated with one another. Although such work documents these associations, it stops short of elaborating on the channels through which the two crises could be related.

Theoretical literature provides some further insight into how currency crises and debt crises are connected, however no empirical tests have been done to suggest which theories most plausibly hold in practice. My paper thus seeks to contribute to existing literature by attempting to find empirical support for some transmission channels that have been put forward in theory to explain the link between currency and debt crises. Below, I outline the theoretical evidence for how currency and debt crises could be

[1] In these papers, the percentage of debt crises associated with currency crises in the sample as a whole is defined as the number of years for which both a currency crisis and a debt crisis occurred, divided by the total number of years for which a debt crisis occurred. Similarly, the percentage of currency crises associated with debt crises is defined as the number of years for which both crises occurred, divided by the total number of years for which a currency crisis occurred.

related, categorizing potential links into common causes, currency and debt crises as substitutes, transmission from debt crisis to currency crisis, and transmission from currency crisis to debt crisis.

Common Causes: Many studies provide evidence for how currency and debt crises can both be independently caused by common external macroeconomic variables, such as a recessionary environment, political changes, or institutional or structural issues (Dreher, Herz and Karb 2006). Within literature on leading indicators, many indicators have been found useful in predicting both debt and currency crises, including but not limited to low reserves, low GDP growth, low foreign direct investment, and a high debt burden, which could also be indicative of common causes (Detragiache and Spilimbergo 2001 ; Reinhart 2002 ; Herz and Tong 2004 ; Dreher, Herz and Karb 2006). Indeed, the fact that any crisis is more likely to occur during times when the economy is weak has been well-documented (Reinhart and Rogoff 2009), and so it is unsurprising that episodes of currency crises and episodes of debt crises can have macroeconomic causes in common.

Substitutes: Currency and debt crises are presented as strategic policy substitutes through the government's budget constraint in some theoretical papers. The models consider a welfare-maximizing government with a strained budgetary position needing to finance some debt coming due. Inflating away the value of the debt through monetary expansion (leading to a currency crisis), and refusing debt payments coming due (leading to a debt crisis) enter the models as substitute options for servicing the debt (Calvo 1998 ; Benigno and Missale 2001 ; Bauer, Herz and Karb 2003 ; Jahjah and Montiel 2003 ; Herz and Tong 2004 ; Sandleris 2012). However, these models specifically presuppose that the government faces debt difficulties, and do not consider how external pressure on the currency might affect the policy trade-off.

Transmission from debt to currency: Literature exploring transmission from

debt to currency crises focuses on the role of recessionary conditions and capital outflows as transmission channels. Dreher, Herz and Karb (2006) suggest that debt crises can trigger a capital outflow, which puts pressure on the currency to depreciate. Dooley (2000) and Rose (2002) provide evidence for debt crises being associated with losses in trade, output, and employment, which similarly exacerbate pressure on the currency to depreciate.

Transmission from currency to debt: Literature exploring transmission from currency to debt crises suggests that pressure on debt can build up following a devaluation from higher interest rates, from reputation losses, or from the nominal increase in the value of foreign currency-denominated debt. Flood and Jeanne (2000) investigate how a government's decision to raise interest rates in the wake of a currency crisis has repercussions on the debt burden. Papers looking at currency crises in pegged economies have found that devaluation is linked to reputation losses, downgrades in credit rating, and higher risk premiums, which in turn increase the risk of a debt crisis (Calvo and Reinhart 2000 ; Reinhart 2002).

Finally, many papers have found foreign currency-denominated debt to be a natural link between currency and debt crises. Foreign currency-denominated debt is vulnerable to movement in exchange rates, and a sharp depreciation greatly increases the value of such debt relative to local currency (Eichengreen, Hausmann and Panizza 2002). A country with high levels of foreign currency-denominated debt thus finds itself facing a much higher debt burden following a currency crisis, which increases the likelihood of a debt crisis. Flood and Jeanne (2000) also find that private foreign currency-denominated debt can affect the likelihood of a debt crisis after a currency crisis if firms with significant foreign currency-denominated debt find themselves in financial difficulty. However, other papers find that countries with high levels of foreign currency-denominated debt can avoid joint currency and debt crises if they also

have large foreign currency assets, and consequently a smaller aggregate currency mismatch (Goldstein and Turner 2003 ; Bordo and Meissner 2005 ; Bordo, Meissner, and Stuckler 2009).

Existing literature thus provides different explanations for how currency and debt crises can be connected. The goal of my paper is to investigate whether empirical evidence is consistent with the explanations. In the subsequent section, I build on these theories to propose transmission channels that connect currency and debt crises. This is followed by an empirical analysis of the channels.

3 Transmission Channels

The previous section outlined different theories of how currency and debt crises might be connected. In my paper, I further explore the theories of connections between currency and debt crises via foreign currency-denominated debt to advance some potential transmission channels. I then empirically test these channels later on in the paper.

I choose to focus on foreign currency-denominated debt in particular because of its prominence in many countries today.[2] To illustrate its potential significance to currency and debt crises, I present the average level of foreign currency-denominated debt relative to GDP lagged by one year in my sample in Table 1 below. Public foreign currency-denominated debt to GDP is significantly higher during a crisis than during non-crisis years. It is highest immediately preceding a joint currency and debt crisis at 57% on average, compared to 22% during a currency crisis, 44% during a debt crisis, and 19% during non-crisis years. Although no inferences about causality can be made at this stage, these statistics suggest an association between high levels of foreign currency-denominated debt and joint currency and debt crises. While there is no similarly stark discrepancy for private foreign currency-denominated debt, its prominent role in the Asian crisis of 1997 and its increasing prevalence in many countries today caution against ignoring its role altogether (Corsetti, Pesenti and Roubini 1998 ; Pomerleano 1998 ; Krugman 1999).

These summary statistics thus provide some observational evidence for a connec-

[2]The inability of many countries to issue external debt in their domestic currency is well-documented in literature on 'original sin' (Eichengreen, Ricardo, and Hausmann 2002, 2003). The literature finds that the external debt of emerging and developing countries is overwhelmingly denominated in five major currencies: the US dollar, the euro, the pound sterling, the Swiss franc, and the yen. For example, Eichengreen, Ricardo, and Hausmann (2002) find that 96.6% of outstanding securities placed in international markets between 1999 and 2001 were denominated in one of those five currencies.

tion between foreign currency-denominated debt and joint currency and debt crises. Therefore, my paper focuses on potential transmission channels that operate through this connection.

Table 1: Foreign Currency-Denominated Debt to GDP

	Joint crisis	Currency crisis	Debt crisis	No crisis
Public FC debt/GDP	57.4%	22.1%	44.0%	19.4%
Private FC debt/GDP	3.5%	4.0%	5.2%	5.7%

Notes: Table 1 reports the level of one-year lagged foreign currency-denominated debt relative to GDP for joint currency and debt crises, currency-only crises, debt-only crises, and non-crisis times. Further details on the foreign currency-denominated debt data used are included in the Appendix.

Below, I describe the potential transmission channels investigated in this paper, and the ways in which they could play out in accordance with literature on foreign currency-denominated debt. Independently, the performance of these variables during a currency crisis and their relationship to debt sustainability are well-documented. My contribution therefore lies in combining those two strands of evidence to explain how these channels can connect currency and debt crises. In following sections of my paper, I also analyze whether empirical evidence is consistent with the presence of these transmission channels.

The transmission channels that I test can play out in two main ways. Through the channels described below, a currency crisis (i) depletes foreign currency assets, thereby exacerbating the currency mismatch and making the public foreign currency-denominated debt burden more difficult to maintain; and (ii) adversely impacts the government's fiscal balance, thereby leaving fewer resources to honor its debt commitment (regardless of denomination).

(1) **Foreign exchange reserves**: Foreign exchange reserves are used by central banks to protect the currency against a sharp depreciation, or, once a currency crisis

has begun, to prevent the currency from depreciating further. Currency crises are therefore associated with a sharp decline in reserves (Kaminsky and Reinhart 1999). However, a decline in reserves exacerbates the aggregate currency mismatch, as it represents a decline in foreign currency assets. If the public sector holds substantial foreign currency-denominated debt, a depletion of foreign currency assets makes it more difficult for the government to service this debt, leading to a higher likelihood of a debt crisis.

(2) **Current account balance**: In the immediate aftermath of a currency crisis, the current account balance deteriorates as trade adjustment to the new exchange rate levels happens gradually. Before firms and individuals are able to revise their behaviors in accordance with new exchange rate levels, the rise in the price of imports and the fall in the price of exports cause the current account to deteriorate as imports become relatively more expensive and exports relatively less expensive. Indeed, previous studies have documented that currency crises are associated with both a decline in exports and a decline in imports (Kaminsky, Lizondo and Reinhart 1998 ; Kaminsky and Reinhart 1999). The adverse effect on the current account may be larger if firms with substantial foreign currency-denominated debt are affected more severely by the economic decline due to adverse effects on their balance sheets, and are unable to produce or export as much. The decline in the current account represents a decline in foreign currency assets, and hence exacerbates the currency mismatch, making it more difficult for the government to honor its foreign currency-denominated debt commitments.

(3) **Tax revenue**: Currency crises are associated with a deterioration of the government's fiscal balance (Kaminsky, Lizondo and Reinhart 1998). I expect part of this deterioration to stem from a fall in tax revenue due to the adverse economic conditions resulting from the currency crisis. Furthermore, if firms have significant

foreign currency-denominated debt exposure, the effect of the depreciation on their balance sheets can put them in financial difficulty and result in them paying less taxes as a result (Bordo and Meissner 2005). In turn, falling tax revenue make a debt default more likely because they adversely impact the government's fiscal balance, leaving the government with fewer resources to honor its debt commitments.

(4) **Government expenditure**: Currency crises are associated with an increase in government expenditure via subsidies, social transfers, and bailout costs, as automatic stabilizers respond to the impact of the crisis (Kaminsky, Lizondo and Reinhart 1998). Furthermore, government expenditures will increase if firms with high foreign currency-denominated debt exposures facing financial difficulty require larger government transfers and subsidies. In extreme cases of systematic distress or distress of large, economically significant firms, the government may have to assume liability for their debts and bail them out, increasing expenditure even more. An increase in government expenditure has an adverse effect on the government's fiscal balance, leaving the government with fewer resources to honor its debt commitments, or, in the extreme case where liability is transferred to the public sector, increases the public debt burden overall.

Building on existing literature, I therefore expect that a currency crisis can make a debt crisis more likely through these four transmission channels. In the following sections, I empirically test the transmission channels to determine whether empirical evidence is consistent with these descriptions.

4 Data

Existing literature contains two broad categories of definitions of a currency crisis: definitions that only take exchange rate depreciation into account (Frankel and Rose 1996 ; Reinhart and Rogoff 2009), and definitions that incorporate additional variables such as reserve losses and interest rate hikes (Eichengreen, Rose, and Wyplosz 1996 ; Kaminsky and Reinhart 1999 ; Glick and Hutchinson 2000). I utilize the Reinhart and Rogoff definition (2009), which defines a currency crisis as an annual depreciation of 15% or above. This parsimonious definition focuses only on the resultant depreciation of the currency, making the analysis more straightforward to interpret.

Following Reinhart and Rogoff (2009), I define a debt crisis as the failure to meet a principal or interest payment on the due date or within the specified grace period, including cases when the debt is restructured into terms less favorable than the original. I include both external and domestic debt crises in my sample to adopt a more comprehensive definition of debt, as both varieties of debt are important to overall debt sustainability and the decision to default (Reinhart and Rogoff 2009).

For my empirical analyses, I use a subset of Reinhart and Rogoff's dataset (2009) which documents crisis occurrence for 68 countries in the period 1800-2014 on an annual basis.[3] I use the data for all the countries in the sample, but restrict observations to 1970 onwards due to data availability for other indicators. For any given country-year, the database contains a dummy variable that equals 1 if the country is experiencing a currency (debt) crisis, and 0 otherwise. Importantly, the dummies equal 1 for the entire duration of a crisis, not just the year of incidence. This allows me to capture crises in their entirety, and observe the long-term effects, since both currency and debt crises frequently last for longer than one year. Using dummy vari-

[3]The list of countries in the sample is provided in the Appendix.

ables to proxy for crises has the limitation of not allowing me to observe differences in severity between crises. However, it also allows for more straightforward interpretation, from which research into severity differentials could follow in the future. The database is openly available and has been widely used in crisis analyses.

I obtain variables on real and nominal GDP, inflation, foreign exchange reserves, tax revenue, government expenditure, and the current account balance from the World Bank's Open Data database for the 68 countries included in the Reinhart and Rogoff dataset (2009). The observations are provided on an annual level, allowing me to easily merge the two datasets. More information about these variables is included in the Appendix.

4.1 Summary Statistics

The resultant dataset is an unbalanced panel of 68 countries in the period 1970-2014, with data reported on an annual basis, for a total of 3,060 country-year observations. The dataset contains 266 unique currency crises and 79 unique debt crises. The breakdown of country-year observations by crisis incidence is provided in Table 2 below.

Table 2: Crisis Incidence in Sample by Country-Years

	No debt crisis	Debt crisis	Total
No currency crisis	2,141	329	2,470
Currency crisis	399	191	590
Total	2,540	520	3,060

Notes: Table 2 reports the breakdown of country-year observations by crisis incidence in my sample.

The unconditional probabilities of the crises suggest that debt crises are rarer than currency crises in my sample. I define the unconditional probability of a crisis as the number of years in a crisis divided by the total number of years. For the

sample as a whole, the unconditional probability of a currency crisis is 19%, while the unconditional probability of a debt crisis is 17%.

The conditional probabilities provide preliminary evidence of an association between currency and debt crises. Conditional on a currency crisis, the probability of a debt crisis is 32%, while the probability of a currency crisis conditional on a debt crisis is 37%.[4] Expanding the time window of the association, the probability of having a currency crisis within two years of a debt crisis (with the currency crisis either before or after the debt crisis) is 55%, while the probability of having a debt crisis within two years of a currency crisis is 35%. The conditional probabilities are thus higher than the unconditional probabilities, suggesting that a currency (debt) crisis is more likely given a debt (currency) crisis than unconditionally. These summary statistics are consistent with similar findings reported in existing literature regarding currency and debt crisis frequency and conditional probabilities for different country-year samples (Reinhart 2002 ; Herz and Tong 2004).

[4]Following existing literature described earlier (Reinhart 2002 ; Herz and Tong 2004), I define the probability of a currency crisis conditional on a debt crisis for the sample as the number of years for which both a currency crisis and a debt crisis occurred, divided by the total number of years for which a debt crisis occurred. I define the probability of a debt crisis conditional on a currency crisis for the sample as the number of years for which both crises occurred, divided by the total number of years for which a currency crisis occurred.

5 Empirical Methodology

In my empirical analysis, I aim to answer three questions to explore the nature of transmission between currency and debt crises: (i) whether currency crises and debt crises help predict each other, (ii) whether joint currency and debt crises are more severe than individual crises, and (iii) what some channels are that could help explain their association.

The first part of my analysis follows the methodology of Reinhart and Rogoff (2010a) in using a VAR for currency and debt crisis lags to assess how the crises are associated. The second part of my analysis uses an OLS specification to document differences in real GDP change and inflation associated with currency, debt, and joint currency and debt crises. The methodology is consistent with conventional approaches to studying economic growth (e.g. Husain, Mody and Rogoff 2004 ; Levi-Yeyati and Sturzenegger 2003).

In the third part of my analysis, I use original methodology to test the potential transmission channels presented earlier in this paper. First, I run a regression-based event study analyzing the performance of potential transmission channel variables during a currency crisis. Second, I run a linear probability model measuring the impact of those variables on the likelihood of a debt crisis. Combining the coefficients from the two models, I thereby assess their overall role in the transmission from a currency crisis to a debt crisis. Following the earlier section, the transmission channels tested include foreign exchange reserves, the current account balance, tax revenue, and government expenditure.

5.1 Joint Currency and Debt Crisis Occurrence

I begin by exploring whether currency crises and debt crises can help predict each other. The summary statistics show that joint currency and debt crises are frequent in my sample, however, this step of the analysis takes one step further to quantify the impact, significance, and direction of transmission in joint crisis occurrence. Adapting the methodology of Reinhart and Rogoff (2010a), I use a standard VAR adapted to panel data to estimate whether a currency (debt) crisis can be predicted by its own lags and the lags of a debt (currency) crisis.

I use dummy variables for currency and debt crises that enter the VAR as endogenous variables, since the causality could be going in either direction, taking three-year lags for both.[5] I incorporate real GDP in logarithmic form as an exogenous variable to control for the size of the economy, and hence for the heterogeneity of patterns between advanced, developing, and emerging countries. In this and following models, I cluster errors on a country level.

The two-equation VAR is thus as follows:

$$debt_{it} = \alpha_1 + \sum_k \beta_{1k} curr_{ik} + \sum_k \gamma_{1k} debt_{ik} + \zeta_1 GDP_{it} + \mu_{1i} + \lambda_{1t} + \epsilon_{1it} \tag{1}$$

$$curr_{it} = \alpha_2 + \sum_k \beta_{2k} curr_{ik} + \sum_k \gamma_{2k} debt_{ik} + \zeta_2 GDP_{it} + \mu_{2i} + \lambda_{2t} + \epsilon_{2it} \tag{2}$$

[5]Three-year lags are chosen following existing literature, which uses time windows of between one and three years to observe the full effects attributable to a crisis. For example, Kaminsky and Reinhart (1999) look at the effects of a crisis within two years of its occurrence, Reinhart and Rogoff (2010a) use three-year lags in their VAR testing banking and debt crises, and Herz and Tong (2004) use one-year lags in a similar VAR.

Where $k \in [t-1, t-2, t-3]$ captures three-year lags, *debt* and *curr* are the dummy variables for debt and currency crises respectively, and α_1 and α_2 are constants. μ_1 and μ_2 are country-fixed effects, λ_1 and λ_2 are year-fixed effects, and ϵ_1 and ϵ_2 are the error terms.

Unlike Reinhart and Rogoff (2010a), my dummy indicators take on a value of 1 for any year during which a crisis takes place, not just the year in which it begins. This allows me to capture transmissions taking place at any time during a crisis, since many crises last longer than one year. To control for the bias this imposes, I include entity-fixed effects at a country level, and time-fixed effects on a year level. The fixed effects help control for entity-nonvarying events, such as a crisis affecting an entire region, and also for time-nonvarying effects, such as one crisis playing out over several consecutive years. I use OLS rather than probit or logit to estimate the equations because maximum likelihood estimation techniques do not perform well with fixed effects (Greene 2002 ; Arellano and Hahn 2007).

At this stage, I do not assume a specific direction for the causality, but rather assume that currency and debt crises can affect each other. Indeed, the VAR is set up in a way such that both the currency and debt dummies are endogenous, and so simultaneous causality is assumed. This means that the coefficients cannot be interpreted as probabilities, since doing so would presuppose a causal relationship in one direction. Nevertheless, the significance and signs of the coefficients can show whether one type of crisis helps predict the other type of crisis.

5.2 Joint Currency and Debt Crisis Severity

In the subsequent step of my analysis, I examine the severity of joint currency and debt crises by looking at how they are associated with real GDP and inflation. Twin crises are frequently reported as being more severe than individual crises in existing

literature (Kaminsky and Reinhart 1999 ; Reinhart and Rogoff 2009); however this analysis has not been done for the combination of currency and debt crises specifically. Establishing that joint currency and debt crises are more severe than currency-only or debt-only crises would emphasize the importance of studying how they are connected, thereby setting the stakes for the rest of my paper.

Following conventional approaches to studying economic performance (e.g. Husain, Mody and Rogoff 2004 ; Levi-Yeyati and Sturzenegger 2003), I use the annual change in real GDP and the level of inflation as measures of crisis severity. To observe differences in these variables between crises, I use an OLS model specified as follows:

$$y_{it} = \alpha + \beta_1 joint_{it} + \beta_2 curr_{it} + \beta_3 debt_{it} + \mu_i + \lambda_t + \epsilon_{it} \tag{3}$$

where the dependent variable y is tested for the annual change in real GDP and the level of inflation. α is a constant, μ and λ are the country- and year-fixed effects respectively, and ϵ is the error term.

As above, I use the dummies *curr* and *debt* for currency and debt crises respectively, which take on the value of 1 in any year where the crisis is occurring. I then construct an interaction variable, $joint = curr \times debt$, between these two dummy variables. *joint* therefore takes on the value of 1 in any year for which both a currency and a debt crisis were occurring. The coefficient on *joint* gives the marginal effect of a joint crisis on the dependent variable, in excess of the effect predicted by simply summing the coefficients on *curr* and *debt*. A significant coefficient on this interaction would indicate that a joint crisis is associated with more severe economic outcomes than would be expected by combining the effects associated with individual debt and currency crises. The total effect on the dependent variable associated with a joint crisis is obtained by the sum of coefficients on *curr*, *debt*, and *joint*: $\beta_1 + \beta_2 + \beta_3$.

The specification suffers from simultaneous causality in that the crises could equally be causing or be caused by the GDP change and inflation the model predicts. For example, it may be the case that a debt crisis produces a recessionary environment resulting in a fall in GDP; but it may also be the case that a fall in GDP leads to creditors reevaluating their expectations of a country's solvency and refusing to roll over debt, resulting in a debt crisis. It is reasonable to assume that the bias from the simultaneous causality is positive and hence that the effects are overestimated because the crises and these macroeconomic effects reinforce each other.[6]

Recognizing this, I can interpret the coefficients as upper bounds on the association between crises and their outcomes in terms of GDP and inflation. I also run a robustness check repeating these regressions for one-year lagged values of the crisis dummy variables as the dependent variables to remove some of the simultaneous causality bias. Overall, this analysis highlights the stakes of my paper: if currency and debt crises are associated with more severe macroeconomic outcomes than currency-only or debt-only crises, it is crucial to understand how they are connected.

5.3 Transmission Channels

In the third part of my analysis, I test whether the potential transmission channels of foreign exchange reserves, tax revenue, government expenditure, and the current account, as described earlier, can help explain the transmission between currency and debt crises. I conduct this analysis in two parts. First, I use a regression-based event study to document the evolution of each variable during a currency crisis. Second, I employ a linear probability model to measure the impact of each variable on the

[6]In general, it is found that a fall in GDP and higher inflation make a crisis more likely, while, on the other hand, a crisis can result in a fall in GDP and higher inflation (Kaminsky and Reinhart 1999 ; Reinhart and Rogoff 2009 ; Reinhart and Rogoff 2010b). The crises and these effects are thus positively correlated, and hence the simultaneous causality imposes bias in the same direction.

likelihood of a debt crisis, following the methodology of Bordo, Stuckler, and Meissner (2009). For each variable, I then cross-multiply the coefficients from the two models.

Ideally, this methodology would estimate the total effect on the transmission from a currency crisis to a debt crisis attributable to each specific transmission channel. However, such an interpretation would be biased due to the endogeneity of the crises and the transmission channel variables, which is discussed in detail later. While the estimates cannot cleanly isolate the desired transmission channel effect, they nevertheless provide evidence to support or reject the existence of the transmission channels.

5.3.1 Currency Crisis Event Study

I use an event study analysis within an OLS framework to test the performance of foreign exchange reserves, tax revenue, government expenditure, and the current account balance during a currency crisis. Observational evidence for the evolution of these variables during currency crises has been presented in literature on leading indicators (Kaminsky, Lizondo and Reinhart 1998 ; Kaminsky and Reinhart 1999); however, my methodology takes this evidence a step further by statistically testing the significance of this evolution via an event study. I specify the dummies t_{-2}, t_{-1}, t_0, t_1, and t_2 for each year starting from two years before the year of crisis onset, and ending two years after crisis onset. This enables me to capture a five year window surrounding the currency crisis onset (two years prior, year of onset, two years post), with t_0 being the year of currency crisis onset respectively. I use the annual change in each of my potential transmission channel variables as outcome variables. In this way, I am able to observe their deviation from average levels when a currency crisis happens. The model is specified as follows:

$$y_{it} = \alpha + \beta_1 t_{-2,it} + \beta_2 t_{-1,it} + \beta_3 t_{0,it} + \beta_4 t_{1,it} + \beta_5 t_{2,it} + \mu_i + \lambda_t + \epsilon_{it} \tag{4}$$

Following previous models, I include country-fixed effects (μ) and year-fixed effects (λ). α is a constant and ϵ is the error term. The coefficients β_i can be interpreted as the change in the dependent variable associated with a certain year relative to a currency crisis. The overall change in the outcome variable associated with a currency crisis is determined by the sum of the β-coefficients, $\sum_{i=1}^{5} \beta_i$.

5.3.2 Debt Crisis Likelihood

Following the event study, I set up a linear probability model to test the relationship between the potential transmission channels and the likelihood of a debt crisis. The methodology is similar to Bordo, Meissner and Stuckler (2009). I regress the *debt* dummy on the level of reserves, tax revenue, government expenditure, and the current account balance in logarithmic form to measure the effect of a percentage change in those variables on the likelihood of a debt crisis. I also include a dummy variable for a currency crisis lagged by one year, and control for real GDP. The specification is as follows:

$$debt_{it} = \alpha + \sum_j \beta_j x_{j,it} + \gamma GDP_{it} + curr_{it} + \mu_i + \lambda_t + \epsilon_{it} \tag{5}$$

where x_j are the potential transmission channel variables. α is a constant, μ and λ are the country- and year-fixed effects respectively, and ϵ is the error term. The coefficients on the independent variables can be interpreted as the increase in the likelihood of a debt crisis associated with a 1% increase in each variable, conditional on a currency crisis.

5.3.3 From Currency Crisis to Debt Crisis

For each of the potential transmission channels tested, the event study determines the percentage change in the variable associated with a currency crisis. The linear probability model documents the increase in the likelihood of a debt crisis associated with a 1% change in that variable, conditional on a currency crisis. By multiplying together the coefficients of the models, I can thereby examine the effect of each individual channel in the transmission from a currency crisis to a debt crisis. For instance, suppose that the event study determines that a currency crisis is associated with a 5% decline in reserves, and the linear probability model determines that a 1% decline in reserves is associated with a 0.5% increase in the likelihood of a debt crisis. I would then estimate that the decline in reserves contributes 2.5% ($5 \times 0.5\%$) to the transmission between a currency crisis and a debt crisis.

Formally, I define the full effect of each transmission channel for each tested as follows:

$$transmission_i = \theta_{1i} \times \theta_{2i} \tag{6}$$

where θ_1 is obtained from the event study and θ_2 is obtained from the linear probability model.

In an ideal world, this coefficient could be interpreted as the increase in the likelihood of a debt crisis through a particular transmission channel conditional on a currency crisis. This would be allow me to directly test the transmission channels presented earlier, which predict for transmission specifically from a currency crisis to a debt crisis. However, in practice, simultaneous causality renders such an interpretation inaccurate. In the event study, the evolution of the variables can be an effect of the currency crisis, or an effect of external pressures that also make a currency crisis

or a debt crisis more likely.[7] Similarly, the linear probability model is designed to show the effect of the potential transmission channels on the likelihood of a debt crisis, however, debt-related pressures can be affecting the transmission channels before the crisis even begins.[8] Furthermore, in both cases, external factors such as political changes that influence expectations about a country's solvency affect the likelihoods of both a debt crisis and a currency crisis.[9] Finally, transmission channels are unlikely to be linear and unilateral: existing literature finds that twin crises are particularly vulnerable to vicious cycles whereby the crises reinforce and exacerbate each other (Kaminsky and Reinhart 1999). Therefore, both models suffer from simultaneous causality and omitted variable bias, and combining the event study with the linear probability model cannot isolate transmission from a currency crisis to a debt crisis only.

However, while the *transmission* estimates cannot be interpreted as the precise increase in the likelihood of a debt crisis after a currency crisis through particular transmission channels, they can nevertheless provide evidence to support or reject the existence of transmission channels in this direction. The simultaneous effects described above reinforce each other, and thus impose a positive bias on the esti-

[7]Consider the example of foreign exchange reserves. In the event study, the fall in reserves can be seen as an effect of the crisis, as reserves are often used to prevent the currency from depreciating further once a currency crisis has hit. However, leading indicators literature also finds that a fall in reserves can be an indicator of an impending currency crisis, as reserves are also often used to try to prevent large depreciation if the currency is under pressure (Kaminsky and Reinhart 1999). Thus a fall in reserves can be both a signal of an impending currency crisis and an effect of a currency crisis. The event study cannot determine the specific effect in each direction.

[8]For instance, a fall in reserves can be precipitated by debt-related pressures. Creditors becoming concerned about the solvency of the country can refuse to roll over debt, which may induce the government to draw on its reserves to service its debt if that debt is denominated in a foreign currency. In this case, the refusal to roll over debt influences both the decline in reserves and the likelihood of a debt crisis.

[9]For instance, suppose a change in leadership at the central bank results in looser monetary policy and a willingness to spend reserves down to a lower level than was previously tolerated. In this case, the cause of the fall in reserves is not caused by currency and debt pressures, however, the fact that reserves have fallen affects both the likelihood of a currency crisis and the likelihood of a debt crisis at the same time.

mates. It is reasonable to assume that the effect of interest is overestimated, and the *transmission* estimates provide at most an upper bound for the effects described in the transmission channels section. Moreover, regardless of the specific numerical value, a positive *transmission* estimate for any given channel is therefore consistent with the existence of a transmission effect from a currency crisis to a debt crisis through this channel. While it cannot exactly quantify that transmission effect, a positive *transmission* estimate provides supporting evidence for that effect.

6 Results

The results are reported following the order of the methodology section. I begin by reporting the results of the VAR looking at whether currency crises and debt crises can help predict each other. Next, I describe the results of the OLS model documenting the associations between joint currency and debt crises and their macroeconomic outcomes in terms of real GDP and inflation. Finally, I provide the results for the event study and linear probability models examining which channels can be helpful in explaining the association between currency and debt crises.

6.1 Joint Currency and Debt Crisis Occurrence

Table 3 reports the output of the panel VAR with currency and debt crisis lags. The columns represent two separate regressions determined simultaneously. The main result of this model is that currency crises help predict debt crises. Debt crises also help predict currency crises, however they do so over a shorter time frame.

The coefficients of interest, reported in the table in bold, are the lags of currency (debt) crises in relation to the likelihood of a debt (currency) crisis. Column (1) shows that the first two lags of a currency crisis are significant in predicting a debt crisis, while the third lag is insignificant. Given the endogeneity inherent to a VAR model, it is inaccurate to interpret the coefficients as probabilities. However, the positive, significant coefficients on the first two lags of a currency crisis suggest that a currency crisis helps predict a debt crisis.

On the other hand, column (2) shows that only the first lag of a debt crisis is significant in predicting a currency crisis. This suggests that only a debt crisis in the immediate past can help predict a contemporaneous currency crisis, while a currency crisis helps predict a debt crisis over a slightly longer period of time. Therefore, while

Table 3: VAR of Currency Crises and Debt Crises

VARIABLES	(1) Debt crisis	(2) Currency crisis
Currency crisis (t-1)	**0.0558*** **	0.262***
	(0.0169)	(0.0337)
Currency crisis (t-2)	**0.0407** **	0.150***
	(0.0172)	(0.0249)
Currency crisis (t-3)	**-0.0133**	0.130***
	(0.0141)	(0.0229)
Debt crisis (t-1)	0.601***	**0.102*** **
	(0.0454)	(0.0301)
Debt crisis (t-2)	0.164***	**-0.00199**
	(0.0452)	(0.0327)
Debt crisis (t-3)	0.0551	**0.0376**
	(0.0376)	(0.0339)
Log real GDP	-0.0187**	-0.0467***
	(0.00796)	(0.0159)
Observations	2,687	2,687
Country FE	YES	YES
Year FE	YES	YES

Standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Notes: Table 3 reports the results of a VAR using currency crisis lags and debt crisis lags as specified in equations (1) and (2). The two columns represent separate regressions determined simultaneously. Standard errors clustered by country are reported in parentheses.

column (1) provides some evidence that a debt crisis help predict a currency crisis, the association in this direction appears to be shorter.

6.2 Joint Currency and Debt Crisis Severity

Table 4 reports the output for real GDP change and inflation for currency-only, debt-only, and joint currency and debt crises. It shows that joint currency and debt crises are associated with a steeper fall in real GDP and higher inflation than either crisis individually, highlighting the importance of my contribution in studying the connections between currency and debt crises.

Column (1) reports the results for real GDP. The coefficient on the interaction is significant at the 5% level and notably larger than the coefficients on *curr* and *debt*, suggesting a much steeper marginal effect when the two crises occur at the same time. The point estimate of the total fall in real GDP associated with a joint crisis, calculated as the sum of the coefficients on the dummy variables, is 3.7%. This estimate is much larger than the 1% and 0.5% associated with currency-only and debt-only crises respectively, and is statistically significant at the 1% level using a joint hypothesis test for linear combinations.

Column (2) reports the results for inflation. The extremely large coefficient on the interaction variable suggests that joint currency and debt crises are associated with some of the most severe hyperinflation episodes. The large coefficient on the currency crisis dummy variable reflects a similar but slightly less exacerbated dynamic for currency-only crises. This in itself suggests that joint currency and debt crises are more severe than currency-only or debt-only crises as they are more prominently associated with severe hyperinflation cases. However, for the purposes of comparing outcomes between crises more generally, hyperinflation episodes skew the results and make comparison with more moderate inflation cases difficult.

Table 4: Real GDP Change and Inflation during Crises

VARIABLES	(1) Δ real GDP	(2) Inflation	(3) Inflation < 40%
Joint curr x debt	-0.0212**	175.3*	2.328
	(0.00868)	(95.29)	(1.879)
Currency crisis	-0.0104***	48.36	4.369***
	(0.00309)	(40.71)	(0.690)
Debt crisis	-0.00549	-13.66	2.858**
	(0.00532)	(24.13)	(1.088)
Constant	0.0405***	14.14***	7.188***
	(0.000879)	(4.982)	(0.147)
Observations	2,870	2,747	2,563
R-squared	0.278	0.102	0.547
Country FE	YES	YES	YES
Year FE	YES	YES	YES

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Notes: Table 4 reports the performance of real GDP change and inflation during currency-only, debt-only, and joint currency and debt crises using an OLS model as specified in equation (3). The variable *joint* is an interaction term between the currency crisis and debt crisis dummy variables. Standard errors clustered by country are reported in parentheses.

For this reason, column (3) reports results for the same regression with hyperinflation cases excluded.[10] The coefficient on the interaction variable is no longer significant. This suggests that having both a currency and a debt crisis at the same time does not produce a significant marginal effect on inflation. The coefficients on

[10]For the purposes of this analysis, I define hyperinflation as any episode where annual inflation is equal to or exceeds 40%. The conventional definition of hyperinflation applies the 40% threshold to monthly inflation (Reinhart and Rogoff 2009), however my data is limited to annual observations. Although monthly inflation is a more sensitive indicator of the acceleration of inflation, this definition suffices for removing cases of very high inflation from the sample to allow for a comparison of more moderate values.

curr and *debt* are both positive and significant, confirming existing results regarding the association between inflation and these crises (Kaminsky and Reinhart 1999). The overall increase in inflation associated with a joint currency and debt crisis, as determined by a test of the linear combination of coefficients on the dummy variables, is 9.6%. This point estimate is significant at the 1% level. Therefore a currency and debt crisis is associated with higher inflation than either currency-only or debt-only crises, although the marginal effect of a joint crisis is not significant.

The simultaneous causality bias discussed in the methodology section cautions against interpreting these point estimates as the exact effect, suggesting it is more valid to think of them as upper bounds. Nevertheless, these estimates are useful for comparing the outcomes associated with currency-only, debt-only, and currency and debt crises. The robustness check reported in Table 10 in the Appendix uses one-year lags of the crisis dummy variables as independent variables to remove some of the simultaneous causality. The robustness check supports the main results, that joint currency and debt crises are associated with a larger real GDP decline and higher inflation than either crisis individually.

The model thus provides evidence that joint currency and debt crises are associated with more severe macroeconomic outcomes than either currency-only or debt-only crises. Given these findings, it is extremely important to understand how currency and debt crises are associated. This result therefore demonstrates the importance of studying the connections between currency and debt crises and thereby sets the stakes for my paper.

6.3 Transmission Channels

6.3.1 Currency Crisis Event Study

Table 5 reports the results from the regression-based event study of the evolution of foreign exchange reserves, tax revenue, government expenditure, and the current account balance during a currency crisis. Overall, the model provides evidence for a decline in foreign exchange reserves, and a decline in tax revenue in the three-year time window surrounding the onset of a currency crisis.

Column (1) shows the evolution of foreign exchange reserves, which, as expected, is negative. The fall in foreign exchange reserves is significant in the year before crisis onset and the year of crisis onset, which is consistent with the interpretation that reserves are used as part of monetary policy to attempt to prevent depreciation immediately before a currency crisis onset. Similarly, column (2) shows a significant decline in tax revenue associated with a currency crisis. The decline in tax revenue is significant at the 1% level in the year of crisis onset and the subsequent year, but is not significant prior to the crisis, suggesting that this decline is a symptom rather than a cause of the crisis-related shock.

Counterintuitively, column (3) shows that the coefficients on government expenditure are also negative and significant during and after crisis onset, but positive and significant before crisis onset. This result is contrary to the theory of automatic stabilizers, according to which government expenditure increases during a crisis due to higher social transfers. However, a reason for this result could be that, consistent with other literature on crises (e.g. Kaminsky and Reinhart 1999 ; Reinhart and Rogoff 2010a), I use the annual change in the variables' values reported in US dollars in this event study. The coefficient thus captures both the real decrease in the variable, and also the effect of the nominal depreciation from the currency crisis. This

Table 5: Regression-based Event Study for a Currency Crisis

VARIABLES	(1) Δ Reserves	(2) Δ Tax revenue	(3) Δ Gov Exp	(4) Δ CA
t_{-2}	-0.0207	-0.00836	0.00171	0.0881
	(0.0271)	(0.0157)	(0.0149)	(0.0883)
t_{-1}	-0.0543**	-0.000489	0.0301**	0.0735
	(0.0251)	(0.0145)	(0.0127)	(0.0686)
t_0	-0.135***	-0.0978***	-0.0806***	0.0293
	(0.0244)	(0.0155)	(0.0172)	(0.0711)
t_1	-0.0703	-0.189***	-0.220***	-0.286
	(0.0597)	(0.0321)	(0.0335)	(0.172)
t_2	-0.0700	-0.0657**	-0.0354	0.229
	(0.0668)	(0.0303)	(0.0354)	(0.163)
Constant	0.167***	0.112***	0.108***	-0.0312*
	(0.00431)	(0.00269)	(0.00272)	(0.0162)
Observations	2,783	1,851	1,805	2,155
R-squared	0.181	0.363	0.331	0.087
Country FE	YES	YES	YES	YES
Year FE	YES	YES	YES	YES

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Notes: Table 5 reports the evolution of foreign exchange reserves, tax revenue, government expenditure, and the current account balance over a five-year time window surrounding the onset of a currency crisis. An OLS event study is used as specified in equation (4). The dependent variables are specified in terms of annual percentage change. The regression coefficients can thus be interpreted as the percentage change in the dependent variable associated with a particular year relative to a currency crisis. The observations are taken on an annual level. The differences in the number of observations across regressions is due to differences in data availability for particular variables. Standard errors clustered by country are reported in parentheses.

is unproblematic for measuring the evolution of foreign exchange reserves and the current account balance, since these variables are denominated in foreign currency and contribute foreign currency assets to the economy. However, it can be argued that this introduces a bias in assessing the evolution of tax revenue and government expenditure, since these variables are denominated in domestic currency and are not predominantly used for foreign currency transactions.

I therefore include a robustness check to determine whether the results hold if I use the changes in tax revenue and government expenditure measured in domestic currency as dependent variables. The results are reported in Table 11 in the Appendix. The decline in tax revenue associated with a currency crisis is significant regardless of whether it is reported in US dollars or in domestic currency, however, the decline in government expenditure is no longer significant if the values are reported in domestic currency. In fact, the coefficients on government expenditure after the currency crisis onset in the robustness check are positive, although not significant. This suggests that the decline in government expenditure in Table 5 is driven by the currency depreciation rather than a real decrease in government expenditure. I therefore interpret this failure to pass the robustness check as evidence that there is no real significant change in government expenditure associated with a currency crisis. Finally, column (5) shows no significant effect on the current account balance.

Overall, I interpret this table as providing evidence for significant declines in foreign exchange reserves and tax revenue during a currency crisis. Given that most of the significant coefficients on these variables fall within one year of crisis onset, I consider the time window $\{t_{-1}, t_0, t_1\}$ to encapsulate the full effects of the currency crisis. Therefore, to obtain the entire effect associated with the currency crisis, I sum the coefficients on those dummies for each variable. This method predicts point estimate declines of 26% in foreign exchange reserves and 29% in tax revenue associated

with a currency crisis.

These point estimates are very large, however it is important to remember that the methodology does not differentiate between crises of different severity. These estimates present average changes associated with crises, which are likely skewed upwards by unusually strong effects of very severe crises. In addition, the simultaneous causality and omitted variable bias associated with this specification have been discussed at length in the methodology section, and suggest that the coefficients should be interpreted as overall associations between a currency crisis and these variables.

6.3.2 Debt Crisis Likelihood

I follow up the transmission channel analysis by using a linear probability model to explore the association between the selected variables and the likelihood of a debt crisis. Table 6 reports the results. I find evidence that, conditional on a currency crisis, a decrease in foreign exchange reserves and a decrease in tax revenue are associated with an increased likelihood of a debt crisis.

Due to the limited number of observations for government expenditure and the current account balance, I include them in separate specifications. The coefficients on foreign exchange reserves and tax revenue are persistently negative and significant across specifications. This suggests an association between falling foreign exchange reserves and an increase in the likelihood of a debt crisis; and also an association between falling tax revenue and an increase in the likelihood of a debt crisis. The coefficients on government expenditure and the current account balance are not significant. My preferred specification is reported in column (1), as it contains the largest number of observations and only includes variables that are most consistently significant. I use this specification as the base for the remainder of my analysis.

Specifically, the regression predicts that a fall in foreign exchange reserves by 1%

Table 6: Linear Probability Model for a Debt Crisis

VARIABLES	(1) Debt crisis	(2) Debt crisis	(3) Debt crisis	(4) Debt crisis
Log foreign exchange reserves	-0.0588***	-0.0582***	-0.0682*	-0.0672*
	(0.0187)	(0.0197)	(0.0390)	(0.0397)
Log tax revenue	-0.128**	-0.0515	-0.200***	-0.0613
	(0.0582)	(0.0837)	(0.0727)	(0.0423)
Log government expenditure		-0.100		-0.181*
		(0.0704)		(0.0946)
Log current account balance			0.0175	0.0147
			(0.0124)	(0.0125)
Log GDP	0.0965	0.113	0.293***	0.412***
	(0.0802)	(0.0858)	(0.0993)	(0.107)
Currency crisis	0.108***	0.0982***	0.173***	0.163**
	(0.0299)	(0.0309)	(0.0598)	(0.0627)
Constant	1.957	2.058	-1.644	-3.751**
	(1.691)	(1.773)	(2.400)	(1.822)
Observations	1,851	1,820	556	548
R-squared	0.508	0.511	0.670	0.659
Country FE	YES	YES	YES	YES
Year FE	YES	YES	YES	YES

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Notes: Table 6 reports a linear probability model for the likelihood of a debt crisis conditional on changes in the foreign exchange reserves, tax revenue, government expenditure, and the current account balance respectively as specified in equation (5). The dependent variable is a dummy variable indicating the incidence of a debt crisis, while the independent variables are reported in logarithmic form. The coefficients can therefore be interpreted as the percentage change in the likelihood of a debt crisis associated with a percentage change in the independent variables. Due to data availability, government expenditure and the current account balance are reported in separate specifications. The differing number of observations across specifications is likewise due to data availability for particular variables. My preferred specification is reported in column (1). Standard errors clustered by country are reported in parentheses.

increases the likelihood of a debt crisis by a small but statistically significant amount (the point estimate in my preferred specification, rounded to one significant figure, is 0.06%). The same is true for a fall in tax revenue, with a point estimate of 0.1%.

The use of a linear probability model presents limitations for the interpretation of the coefficients as probabilities as it assumes a linear relationship between the dependent and independent variables. In this case, it is unlikely that foreign exchange reserves and tax revenue have a linear relationship with the likelihood of a debt crisis. The relationship likely depends on the magnitude of the decline. Small declines in foreign exchange reserves or tax revenue may be perceived as part of the economic cycle and thus have smaller repercussions on debt crisis likelihood than predicted by this model, while large declines may be seen as effects or manifestations of severe economic issues and have a much stronger effect on debt crisis likelihood. Moreover, the effect of foreign exchange reserves and tax revenue on the likelihood of a debt crisis will depend on their initial level. For example, a decline in foreign exchange reserves will present fewer concerns for an economy with high levels of foreign exchange reserves than one with low levels of foreign exchange reserves.

The coefficients reported in this model therefore represent average probabilities along both these dimensions. Following the discussion of simultaneous causality and omitted variable bias in the methodology section, the coefficients should be interpreted as overall associations between the variables and debt crisis likelihood.

6.3.3 From Currency Crisis to Debt Crisis

To compare the effects of the transmission channels, I combine the results from the regression-based event study with the results from the linear probability model. Following equation (6), I multiply the aggregate coefficients from the event study, defined as the sum of the coefficients on dummies t_{-1}, t_0, and t_1, by the coefficients from the

linear probability model. I do this exercise for foreign exchange reserves and tax revenue, as those are the only variables significant in both models. The results provide evidence supporting that foreign exchange reserves and tax revenue can help explain the connection between currency and debt crises in the ways elaborated in transmission channels section. Table 7 presents the estimates of the transmission coefficients.

Table 7: Transmission Estimates

Channel	θ_1	θ_2	**transmission**
Reserves	0.26	0.06	**0.016**
Tax revenue	0.29	0.1	**0.029**

Notes: Following equation (6), θ_1 is taken from the currency crisis event study while θ_2 is taken from the debt crisis linear probability model. Specifically, θ_1 is the sum of the coefficients on $\{t_{-1}, t_0, t_1\}$ as determined in the currency crisis event study reported in Table (5). θ_2 is the coefficient from my preferred specification of the linear probability model reported in column (1) of Table (6), rounded to one significant figure.

This exercise estimates that the fall in foreign exchange reserves associated with a currency crisis increases the likelihood of a debt crisis by up to 1.6%, while the decline in tax revenue associated with a currency crisis increases the likelihood of a debt crisis by up to 2.9%. Although modest at first glance, these estimates are non-trivial in comparison to the unconditional likelihood of a debt crisis in my sample, reported in the summary statistics as 17%. The estimates cannot accurately identify a point estimate of how much each channel contributes to transmission from a currency crisis to a debt crisis due to simultaneous causality and omitted variable bias as discussed in the methodology. However, it can reasonably be assumed that the simultaneous effects reinforce each other, allowing for an interpretation of the estimates as upper bounds. The positive values are thus consistent with the transmission channels through reserves and tax revenue described earlier.

Therefore I interpret these estimates as evidence to support that a fall in reserves and a fall in tax revenue associated with a currency crisis increase the likelihood of a debt crisis. The analysis finds no empirical evidence for similar transmission channels via government expenditure or the current account balance. Overall, I therefore find empirical evidence consistent with two of the four transmission channels I set out to test.

7 Conclusion

Despite the fact that joint currency and debt crises are a relatively frequent phenomenon, few papers examine how they are connected. In this paper, I investigate the connection between currency crises by empirically assessing (i) whether currency and debt crises help predict each other, (ii) whether joint currency and debt crises are more severe than currency-only and debt-only crises, and (iii) what some channels are that help explain their connection. I find that currency crises help predict debt crises, while debt crises predict currency crises over a shorter time frame. Consistent with other literature on twin crises, I find that joint currency and debt crises are more severe than currency-only and debt-only crises.

Using a novel methodology combining a regression-based event study with a linear probability model, I find empirical evidence consistent with two of the four transmission channels between currency and debt crises that I investigate. Specifically, I find that the decline in tax revenue following a currency crisis increases the likelihood of a debt crisis by up to 2.9%, while the decline in foreign exchange reserves following a currency crisis increases the likelihood of a debt crisis by up to 1.6%. I find no empirical support for a similar transmission via government expenditure or via the current account balance.

The exact magnitude of the transmission from a currency crisis to a debt crisis via each variable cannot be completely identified. However, these estimates provide useful upper bounds and an indication of which channels are important, from which further work to quantify the exact transmission effects can follow in the future.

In this way, my paper sheds light on the connections between currency and debt crises, which have largely been missing from twin crisis literature. Given the adverse outcomes that such joint crises can produce, understanding these connections is cru-

cial. It is my hope that future research in this field will further recover the 'missing twins' in the literature, and inspire policy to make them less frequent in economies going forward.

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A Appendix

A.1 Countries in Sample

Table 8: Countries in Sample

Countries			
Algeria	Ecuador	Kenya	Portugal
Angola	Egypt	Korea	Romania
Argentina	El Salvador	Malaysia	Russia
Australia	Finland	Mauritius	Singapore
Austria	France	Mexico	South Africa
Belgium	Germany	Morocco	Spain
Bolivia	Ghana	Myanmar	Sri Lanka
Brazil	Greece	Netherlands	Sweden
Canada	Guatemala	New Zealand	Thailand
Central African Republic	Honduras	Nicaragua	Tunisia
Chile	Hong Kong	Nigeria	Turkey
China	Hungary	Norway	United Kingdom
Colombia	Iceland	Panama	United States
Costa Rica	India	Paraguay	Uruguay
Côte d'Ivoire	Indonesia	Peru	Venezuela
Denmark	Italy	Philippines	Zambia
Dominican Republic	Japan	Poland	Zimbabwe

Notes: Table 8 presents the list of countries used in my sample.

A.2 World Bank Open Data Variables

Table 9: Variables used from the World Bank's Open Data database

Variable	Indicator Code	Definition
Real GDP	NY.GDP.MKTP.KD	GDP, constant 2010 USD
Nominal GDP	NY.GDP.MKTP.CD	GDP, current USD
Inflation	FP.CPI.TOTL.ZG	Inflation, consumer prices, %
Reserves	FI.RES.TOTL.CD	Total reserves, current USD
Current account balance	BN.CAB.XOKA.CD	Current account balance, current USD
	BN.CAB.XOKA.GD.ZS	Current account balance, % GDP
Tax revenue	GC.TAX.TOTL.GD.ZS	Tax revenue, % GDP
	GC.TAX.TOTL.CN	Tax revenue, current local currency unit
Government expenditure	NE.CON.GOVT.CD	General government final consumption expenditure, current USD
	NE.CON.GOVT.ZS	General government final consumption expenditure, % GDP
	NE.CON.GOVT.CN	General government final consumption expenditure, current local currency unit

Notes: Table 9 presents the list of variables used in my analyses that are taken from the World Bank's Open Data database. All variables are provided at an annual level.

A.3 Foreign Currency-Denominated Debt Data

I obtain public debt data from the World Bank's World Development Indicators database. Specifically, I exploit the indicator for public and publicly guaranteed (PPG) external debt stocks, outstanding and disbursed (DT.DOD.DPPG.CD). This source contains data on the long-term (>1 year) external obligations of public debtors, including the national government, political subdivisions and their agencies, and autonomous public bodies, as well as external obligations of private debtors that are guaranteed for repayment by a public entity. The data is available on an annual basis from 1970 onwards. It is important to note that this indicator does not encompass domestic debt. Given that domestic debt is predominantly denominated in domestic currency in most countries, while the opposite is true for external debt (Reinhart and Rogoff 2009), this indicator is nevertheless useful for the purposes of my paper.

To obtain the percentage of this public external debt denominated in a foreign currency, I use indicators from the World Bank's International Debt Statistics database, which contains a breakdown of external long-term PPG debt by currency composition. Specifically, it reports the percentage of external long-term PPG debt denominated in Deutschmark, Euro, French franc, Swiss franc, yen, pound sterling, US dollar, special drawing rights, multiple currencies, and all other currencies. I assume the 'all other currencies' proportion is comprised of domestic currency-denominated debt, which is an assumption consistent with previous literature and observable debt patterns (e.g. Bordo, Meissner and Stuckler 2009). To obtain the percentage of PPG debt denominated in a foreign currency for a particular country at a particular year, I sum up the percentages pertaining to debt denominated in Deutschmark, Euro, French franc, Swiss franc, yen, pound sterling, and US dollar. While the Deutschmark and French franc are no longer in use, a substantial part of my sample covers a time before the introduction of the Euro (introduced in 2002), and as such represent natural parallels

of hard currencies. Consequently I exclude the countries for which such debt would be domestic currency-denominated debt whenever I use this indicator for my analysis. These excluded countries comprise of the US, Great Britain, Switzerland, Japan, and the Eurozone countries.[11]

Therefore, the proportion of long-term external debt denominated in a foreign currency is defined as follows:

$$\pi_{FC,it} = \pi_{DM,it} + \pi_{EUR,it} + \pi_{FF,it} + \pi_{CHF,it} + \pi_{JPY,it} + \pi_{GBP,it} + \pi_{USD,it} \tag{7}$$

where the π_is are proportions of debt obligations denominated in the relevant currency.

To obtain the stock of debt denominated in a foreign currency, I multiply the total proportion of debt denominated in a foreign currency by the external public and publicly guaranteed debt stock as follows:

$$FCdebtPPG_{it} = \pi_{FC,it} \times ExtdebtPPG_{it} \tag{8}$$

Similarly to public debt, I obtain data on the stock of private debt from the World Bank's World Development Indicators database, specifically the indicator of private nonguaranteed (PNG) external debt stocks (DT.DOD.DPNG.CD). This contains information for the long-term external obligations of private debtors that are not guaranteed for repayment by a public entity, which mostly consists of debt held by financial and nonfinancial corporations, private not-for-profits, and private individuals or households.

[11]For consistency, I exclude all countries which have adopted the Euro as of January 1 2019. These countries in my sample are Austria, Belgium, Finland, France, Germany, Greece, Italy, Netherlands, Portugal, and Spain.

Data on the currency composition of private debt is not as readily available. I obtain two proxies available for my use. First, following the World Bank's own estimation procedures (International Debt Statistics 2017), I can assume that the currency composition of PNG debt mirrors that of PPG debt. This assumption is warranted if we assume that debt markets for the public and private sectors within a country function in similar ways that might induce public and private entities to have a similar proportion of their long-term borrowing in a foreign currency. Second, I obtain the proportion of debt securities denominated in a foreign currency specifically for corporations from the Bank of International Settlements' Debt Securities Statistics database.

The latter indicator has two drawbacks. First, it is only available for 2003 onwards, which significantly restricts the number of observations I can use. Second, it only encompasses information on debt securities, which notably excludes bank lending, swaps, and other financial instruments that make up significant proportions of corporate balance sheets.

On comparing the two proxies for countries and years when they are both available, the results are remarkably similar. I therefore select to use the proportion of PPG debt denominated in a foreign currency as described above to proxy for the same proportion for PNG debt. As above, I obtain the total stock of external debt denominated in a foreign currency by multiplying this proportion by the stock of PNG debt outstanding and distributed.

A.4 Robustness Check 1: Joint Crisis Severity

Table 10: Joint Crisis Severity with One-Year Lags

VARIABLES	(1) Δ Real GDP	(2) Inflation	(3) Inflation < 40%
Joint crisis	-0.00556	310.9**	2.865
	(0.00504)	(129.2)	(1.916)
Currency crisis	-0.0111***	7.315	5.851***
	(0.00276)	(7.583)	(0.709)
Debt crisis	-0.00584	-8.889	2.349**
	(0.00421)	(16.00)	(0.898)
Constant	0.0397***	13.39**	7.092***
	(0.000791)	(5.466)	(0.146)
Observations	2,870	2,696	2,512
R-squared	0.259	0.122	0.580
Country FE	YES	YES	YES
Year FE	YES	YES	YES

*** p<0.01, ** p<0.05, * p<0.1

Notes: Table 10 presents the results of equation (3) which investigates the differences in real GDP change and inflation between currency-only, debt-only, and joint currency and debt crises. The crisis dummy variables enter these regressions as one-year lagged values for a robustness check of the main analysis presented in Table 4. Standard errors clustered by country are reported in parentheses.

A.5 Currency Crisis Event Study: Graphical Interpretation

Figure 1: Evolution of selected variables during a currency crisis

Notes: Figure 1 presents a graphical interpretation of the currency crisis event study reported in Table 5. The plots present the annual change in foreign currency reserves, tax revenue, government expenditure, and current account respectively in a five-year time window surrounding the onset of a currency crisis. 95% confidence intervals are also depicted.

A.6 Robustness Check 2: Currency Crisis Event Study

Table 11: Currency Crisis Event Study, reported using Domestic Currency

VARIABLES	(1) Δ Tax Revenue	(2) Δ Government Expenditure
t_{-2}	-0.0630***	-0.0439**
	(0.0179)	(0.0168)
t_{-1}	-0.0440**	-0.00285
	(0.0197)	(0.0170)
t_0	-0.0532**	-0.0100
	(0.0210)	(0.0199)
t_1	0.0906*	0.0571
	(0.0486)	(0.0455)
t_2	0.130	0.0376
	(0.0985)	(0.0477)
Constant	0.174***	0.169***
	(0.00470)	(0.00464)
Observations	1,864	2,712
R-squared	0.394	0.346
Country FE	YES	YES
Year FE	YES	YES

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Notes: Table 11 presents the results of equation (4) specifically for the annual change in tax revenue and the annual change in government expenditure, both reported in domestic currency. This serves as a robustness check for the results of the currency crisis event study in the main analysis reported in Table 5. Standard errors clustered by country are reported in parentheses.